

Faucet Actuator

Timed 5-Gallon Bottle Filler

Version 1.0 - July 27, 2026



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Section 1 Project Overview

1.1 Background

My house uses a reverse osmosis filter to obtain clean drinking water. We prefer cool water, so we fill up reusable 5-gallon jugs with the water in order to supply the water cooler. The faucet, however, does not fill up the bottle in one go, as the decrease in pressure is a function of the time that the faucet is open. The angle of the faucet and allowance that the sink provides makes it so that the bottle by itself can only be filled for around a minute before the flow is too low to enter the bottle. My family's solution was to put clothes under the bottle to angle it, thus capturing a wider range of flow rates, decreasing the amount of cycles the faucet must undergo to fill up the bottle.

The faucet still must undergo approximately 3 cycles before the bottle is full, requiring a member to repeatedly turn on and off the faucet to fill the bottle, which requires a small amount of mental energy focused on the task of filling the bottle. If the faucet is not turned off, then the pressure will continually decrease, and the water will drip not into the bottle, as the range of flow rates the bottle can capture does not include extremely low rates, as the bottle opening is not wide enough.



Figure 1: Current faucet setup.

1.2 Project Objective

My goal is to create a device, which, at the press of a button would allow for the bottle to be filled up without the need of supervision. This way, water won't be wasted, and essentially no mental bandwidth would have to be spent on this trivial task.

1.3 Project Timeline

This project took me about 2 weeks to complete, from idea to implementation. There is still a significant amount that can be improved, such as the implementation of perfboard, or even PCB, but it functions well, and further development would yield diminishing returns, as I could spend the time developing other projects.

1.4 Final Outcome

The finished product is a success, with the total time of the filling process taking a little under 1 hour. The bottle fills to about 95% currently, as I wanted to give a little room for a member of my family to fill it up to the desired amount. The total cycles it undergoes is 5, with essentially no water spillage.

Section 2 Engineering Requirements

2.1 Functional Requirements

The system needed to fill the bottle to the desired amount without spilling, needing supervision, or damaging the current system.

2.2 Design Constraints

The current system was not allowed to be altered, and the servo and breadboard sizes must fit within the desired area. The bottle must also be propped up, without any interferences. The setup must be reliable and repeatable.

2.3 Success Criteria

The system must fill up the bottle in less than 2 hours and with fewer than 10 cycles. Full actuation of the faucet must be performed for each cycle.

Section 3 System Overview

3.1 System Architecture

The setup includes an incline to maximize the flow rate range compatible with the bottle opening, a servo, 3D printed components for the servo and faucet, an ESP32, a button, an LED, and two external power sources.

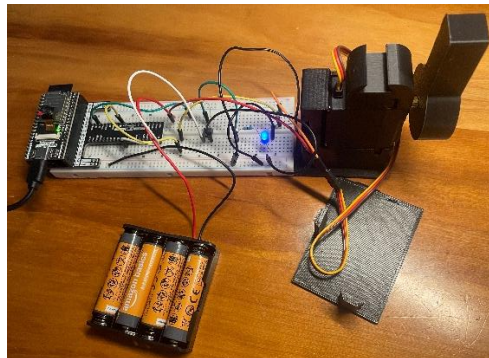


Figure 2: Complete actuation setup.

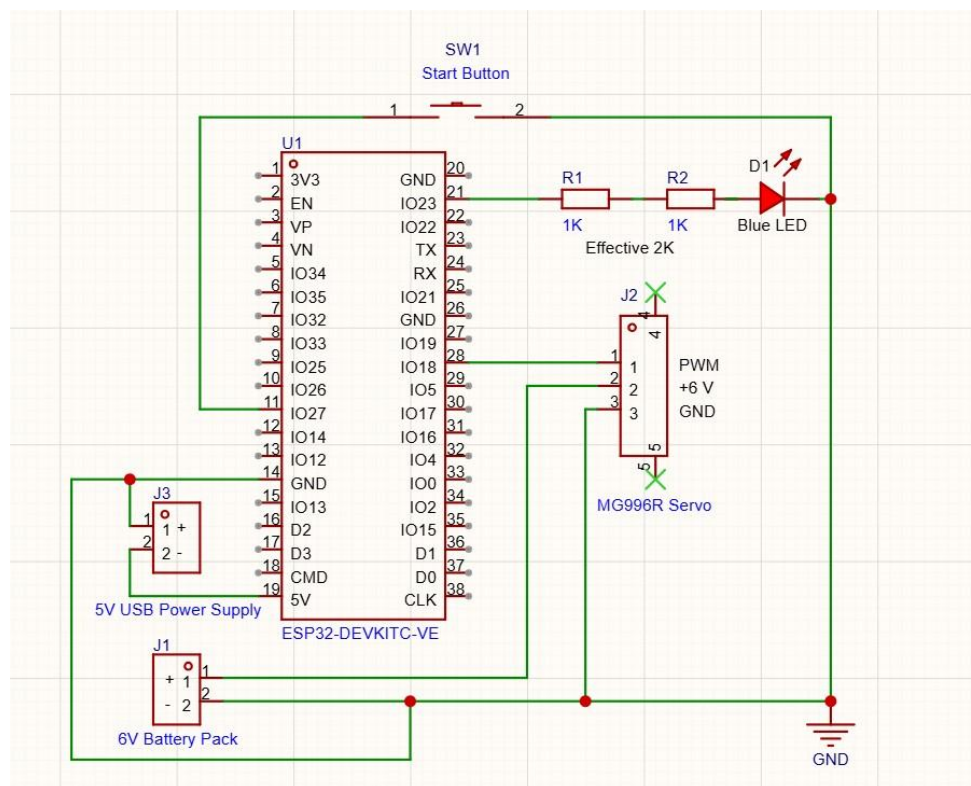


Figure 3: Actuator schematic.

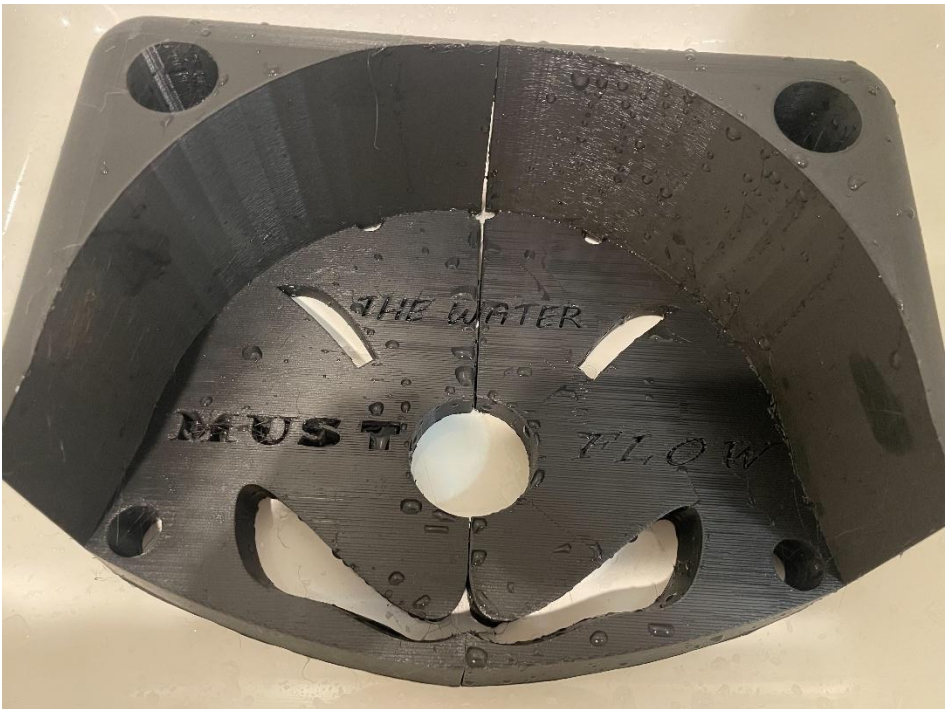


Figure 4: Incline for bottle.

3.3 Bill of Materials

Item	Qty	Part	Source	Unit Cost	Purpose
1	1	MG996R Servo Motor	Amazon	\$5.00	Actuates the RO faucet
2	1	ESP32 Development Board	Amazon	\$8.00	Logic and power source
3	1	6V Battery Pack	Amazon	\$3.00	External Power Source
4	4	AA Batteries	Amazon	\$2.00	External Power Source

Item	Qty	Part	Source	Unit Cost	Purpose
5	1	Button	Amazon Kit	<\$0.10	User Cycle Start
6	1	LED	Amazon Kit	<\$0.10	Cycle Indicator
7	2	1K Ω Resistor	Amazon Kit	<\$0.10	LED Dimmer
8	1	Breadboard	Amazon Kit	<\$1.00	Component Connection
9	10	Wires	Amazon Kit	<\$0.10	Component Connectors

3D-Printed Components

Part	Qty	Material	Print Time (each)	Notes
Servo Mount	1	PLA	3 hr	Mounts servo to bottle stand
Faucet Linkage	1	PLA	1 hr	Connects horn to faucet
Bottle Mount (Half)	2	PLA	24 hr	Holds actuator in position

Section 4 Incline Structure

4.1 Design Development

The design was informed by measurements gathered by me. I then modeled the entire system within SolidWorks, as well as the flow ranges.

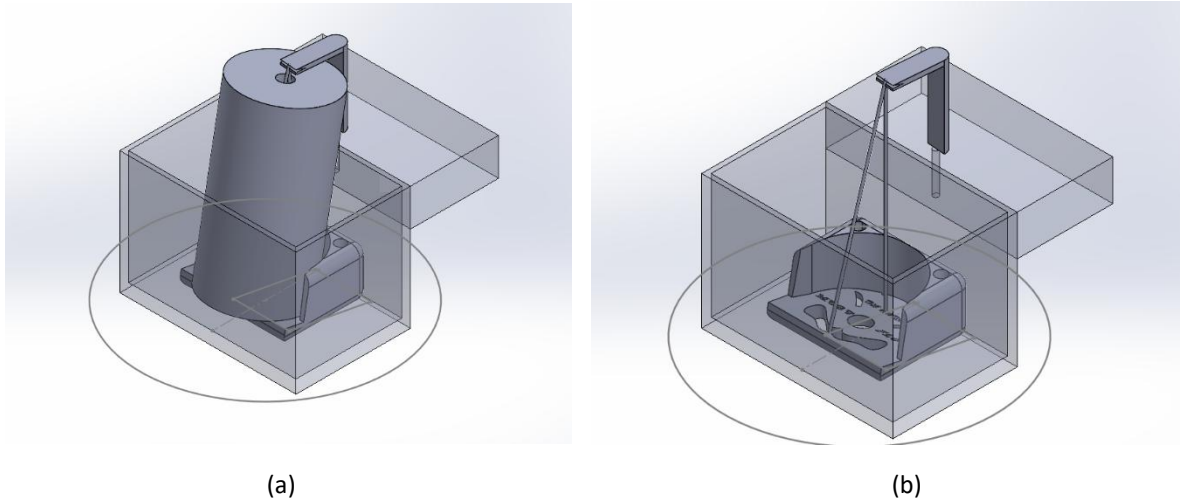


Figure 5: Sink model in SolidWorks based on measurements gathered by me. (a) Shows a simplistic bottle design, with an accurate hole size at the top, ensuring the angle and height of the incline allows for maximum water collection for the pressure range. (b) Shows the methodology for modeling the pressure angle difference, based on flow rate.

To simultaneously ensure that the bottle had no interferences, while maximizing the flow rate range it was compatible with, moving the bottle with the triad function was utilized until an optimal position was found. A cube was then intersected with the bottle position, mounted to the base of the sink, and cut with a cavity to determine the necessary angle of the incline.

4.2 Design Logic

The structure was necessarily going to be large, so it was impossible to make the entire incline one piece. I decided to cut it and print it in halves. I chose not to incorporate fixture methods, as a gap between the two halves allowed for flexibility in bottle size.

I attempted to save material by cutting holes and designs onto the part, as I believed they would reduce the mass and time taken to manufacture the part.

After printing both halves, I compared the time and material to a non cut part, and discovered that I added hours and material to my part, as cutting holes creates more surface area, where walls, which are time consuming and costly, are printed.

Additionally, after testing, I discovered that my measurements were slightly off, and that the incline did not allow for maximum water capture. I designed an attachment to be put on the

incline to increase the incline angle, however I modeled it on a legacy part, meaning my part was not compatible. I decided to solve the issue with software.

4.3 Final Structure



Figure 6: The base is the incline, and the black object to the right is the extra incline I designed, which was not used. It was also modeled before I knew about material savings, so it has a lot of holes and slots, which actually increases the mass and print time.

4.4 Manufacturing

Each print took around 24 hours, using up a little over 300 grams of filament each. There was some stringing in the prints, indicating that the PLA was either not completely dry, the temperature was too high, or that there was some insufficiencies with the nozzle. Regardless, the print quality was sufficient, but I will investigate the cause of stringing.

Section 5 Servo Holder and Actuator

5.1 Design Development

Each component in the servo assembly was measured by me. I first measured the faucet arm's dimensions and vertical displacement, which allowed me to create a mount for it. The servo was measured and modeled by me, so I made the center of rotation concentric to the actuator, allowing for the determination of the necessary servo height. I then did a cavity cut with an intersecting cube and added 0.2 mm of space for printer tolerance, allowing for a press fit. In the model, I mismeasured the servo by .5 cm, which is why the model has additional room on the bottom, as my first print was too small. My second print fit perfectly.

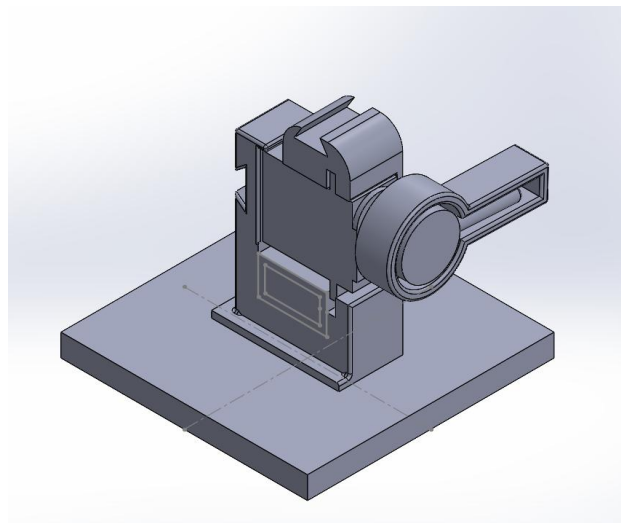


Figure 7: Servo mount and actuator mount assembly.

5.2 Design Logic

The mount has a hole for the servo, with an extra 10 mm for depth. The hole in the top left is for the connecting wires, and the dovetails are for an attachment that allows for the servo to be covered. The extrusion on the bottom left is to counter the axial force induced by CW rotation

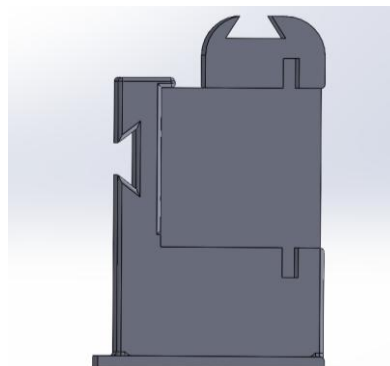


Figure 8: Servo Mount

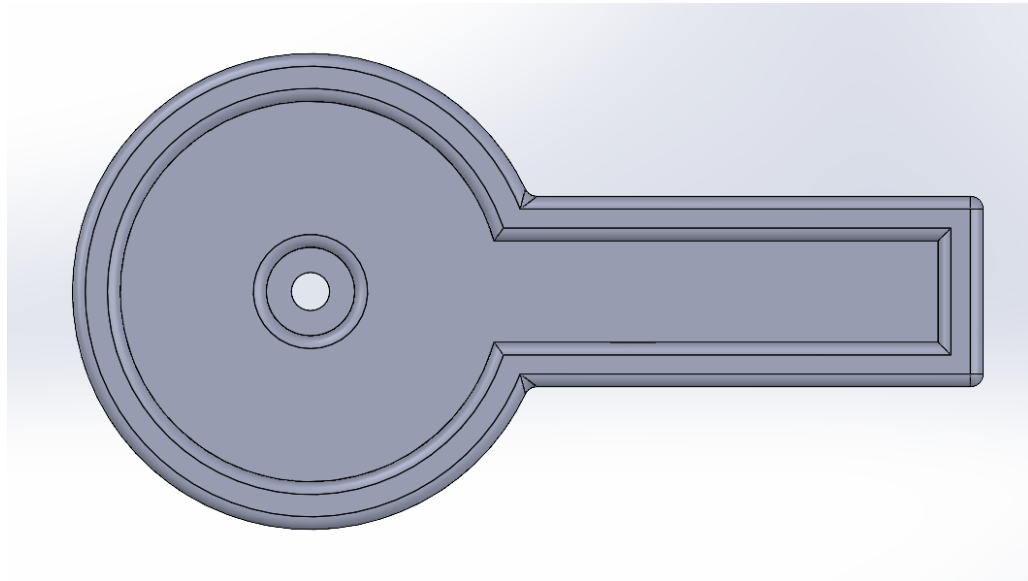


Figure 9: Actuator link. Center extrusion for screw head.

5.3 Final Structure

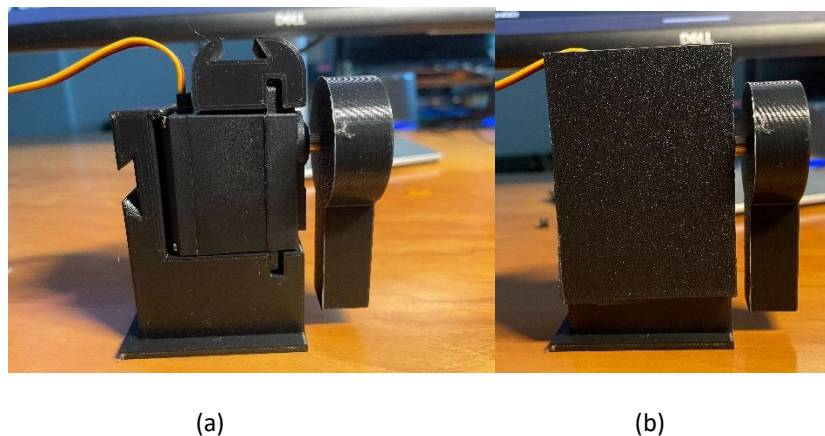


Figure 10: (a) Servo system without cover. (b) Servo system with cover.

5.4 Manufacturing

The print was extremely easy, and I designed all of these parts to be able to be printed with no supports. The cover, however, was printed such that the extrusions are printed vertically. It means that moving the cover laterally could induce enough stress to potentially break it. My initial design for the servo mount incorporated 3 dovetails, which went down to 2 for ease of design. The 3rd dovetail was unnecessary as well. The cover was printed for the previous design, so I utilized my knowledge of print direction to easily break off the extra extrusion. This means that the extrusions are extremely weak, but it is also a non essential piece. Every other component will experience stress in either an inconsequential direction, or the stress will be insignificant due to geometry.

Section 6 Electronics & Software

6.1 Circuit Design

The design of the circuit is extremely basic, as I am attempting to slowly build my knowledge until I develop the skills to take on more complex projects. The ESP32 connects to 3 things, which are the button, the LED, and the servo. I have resistors so that I don't burn out the LED. See Figure 3 for the schematic.

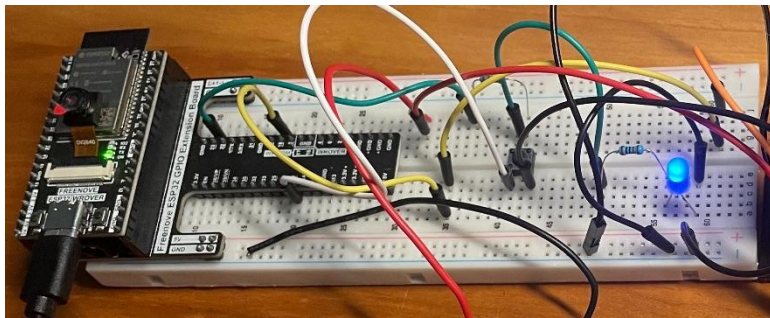


Figure 11: Breadboard setup.

Using a solderful breadboard I got from a hackathon, I soldered the components to the perfboard. Unfortunately, I soldered the ESP32 incorrectly, and the contact was limited. I will have to resolder the ESP32 once my materials arrive.

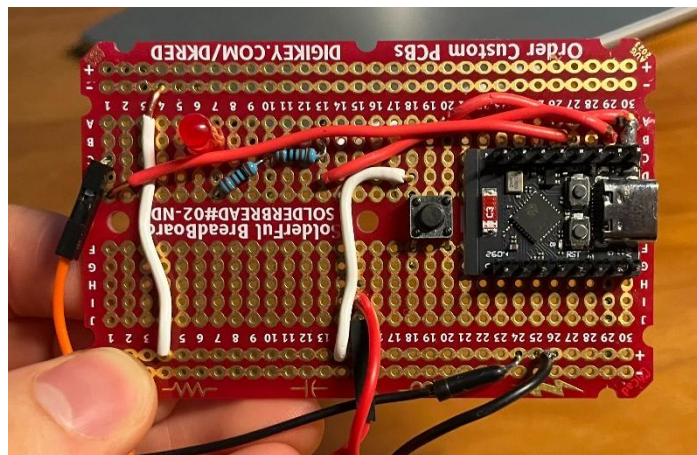


Figure 12: Perfboard setup.

6.2 Embedded Software Logic

I "vibe-coded" this part, but the logic is very simple. I wanted 5 cycles to maximize water pressure, so I have the servo turn at set angles at medium speed. The cycle is structured 110 seconds on, 12 minutes off, allowing for pressure recovery between cycles. The LED blinks to describe which cycle it is on.

Section 7 Data Collection

To determine the dispense and recovery time, I took 10 sample points of filling up 1 Liter of water. The data is as follows:

Cycle	Recharge Time Before Cycle	Open Time Until Flow Slowed (s)
1	Initial (full tank)	23
2	0 s	25
3	0 s	31
4	0 s	35
5	0 s	37
6	0 s	39
7	0 s	45
8	0 s	50
9	3 min	54
10	1 min	56



Figure 13: Data collection

Observing the fall in flow rate, I determined the minimum flow rate, and used the recovery information to map the cycle times.

Section 8 Results

The system functions well, but the axial force pushes back the servo to the point where it is not connected to the faucet. The edge that I designed was not substantial, so the battery pack must be placed directly behind it to counter the induced axial forces. After adjustment, the system functions as intended and fills the bottle to 4.5 gallons in 56 minutes.

Section 9 Future Improvements

I am planning on finishing the perfboard setup, allowing for a more user friendly setup. This will necessitate an enclosure for the perfboard, which will make button presses easier. Additionally, more data collection and analysis can be done for optimal water control, allowing for the minimum cycles and times to fill up the bottle.

I could also incorporate either computer vision or a force sensor for a sort of automated system, but I think my time would be better spent doing other projects.

Section 10 Document Revision History

Version	Date	Author	Change
1.0	7/27/2026	Nathan Robinson	Document Release